```
v = [1 2 3 4]
v =
     1     2     3     4
v = [1; 2; 3; 4;]
v =
     1
     2
     3
     4
v(end-1)
ans =
     3
m = [1 2 3; 6 5 2; 7 5 3]
m =
     1     2     3
     6     5     2
     7     5     3
q = 1.602e-19
q =
   1.6020e-19
a = 6.023e23
a =
   6.0230e+23
mypi = 3.1415
mypi =
    3.1415
numstates = 50
numstates =
    50
year = 2023
```

```
year =
        2023
j2 = [5 6 7 8];
m'
ans =
     1     6     7
     2     5     5
     3     2     3
j = 1:10
j =
     1     2     3     4     5     6     7     8     9    10
j = (1:10)'
j =
     1
     2
     3
     4
     5
     6
     7
     8
     9
    10
j2 = 0:.2:3
j2 =
0    0.2000    0.4000    0.6000    0.8000    1.0000    1.2000    1.4000    1.6000    1.8000    2.0000    2.2000
2.4000 2.6000    2.8000    3.0000
help colon;
m(1,:)
ans =
     1     2     3
m(:,1)
```

```
ans =
    1
    6
    7
m(1,1:2:end)
ans =
    1     3
M = rand(6,6)
M =
   0.8147   0.2785   0.9572   0.7922   0.6787   0.7060
   0.9058   0.5469   0.4854   0.9595   0.7577   0.0318
   0.1270   0.9575   0.8003   0.6557   0.7431   0.2769
   0.9134   0.9649   0.1419   0.0357   0.3922   0.0462
   0.6324   0.1576   0.4218   0.8491   0.6555   0.0971
   0.0975   0.9706   0.9157   0.9340   0.1712   0.8235
M(1:2:end,1)
ans =
   0.8147
   0.1270
   0.6324
M(2:2:end,2)
ans =
   0.5469
   0.9649
   0.9706
M(1:2:end,:)
ans =
   0.8147   0.2785   0.9572   0.7922   0.6787   0.7060
   0.1270   0.9575   0.8003   0.6557   0.7431   0.2769
   0.6324   0.1576   0.4218   0.8491   0.6555   0.0971
```

```
M(1:2:end,1:2:end)
ans =
    0.8147    0.9572    0.6787
    0.1270    0.8003    0.7431
    0.6324    0.4218    0.6555
M(1:2:end,2:2:end)
ans =
    0.2785    0.7922    0.7060
    0.9575    0.6557    0.2769
    0.1576    0.8491    0.0971
zeros(3,4)
ans =
     0     0     0     0
     0     0     0     0
     0     0     0     0
doc zeros
zeros(4)
ans =
     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
A = [1 4; 2 5; 3 6];
sz = size(A);
X = zeros(sz)
X =
     0     0
     0     0
     0     0
f = 7 * ones(5,6)
f =
```

```
     7     7     7     7     7     7
     7     7     7     7     7     7
     7     7     7     7     7     7
     7     7     7     7     7     7
     7     7     7     7     7     7
doc ones
ones(5)
ans =
     1     1     1     1     1
     1     1     1     1     1
     1     1     1     1     1
     1     1     1     1     1
     1     1     1     1     1
x = ones(sz)
x =
     1     1
     1     1
     1     1
rand(2,2)
ans =
    0.6948    0.9502
    0.3171    0.0344
doc rand
r = -5 + (5+5)*rand(10,1)
r =
   -0.6126
   -1.1844
    2.6552
    2.9520
   -3.1313
   -0.1024
```

```
   -0.5441
    1.4631
    2.0936
    2.5469
r = randi([10 50],1,5)
r =
    21    37    36    16    14
randn(2,2)
ans =
   -0.0068   -0.7697
    1.5326    0.3714
doc randn
r = randn(5)
r =
   -0.2256    1.1006   -1.0616    0.8886   -0.1774
    1.1174    1.5442    2.3505   -0.7648   -0.1961
   -1.0891    0.0859   -0.6156   -1.4023    1.4193
    0.0326   -1.4916    0.7481   -1.4224    0.2916
    0.5525   -0.7423   -0.1924    0.4882    0.1978
mu = [1 2];
sigma = [1 0.5; 0.5 2];
R = chol(sigma);
Z = repmat(mu,10,1) + randn(10,2)*R
Z =
    2.5877    6.2141
    0.1955    0.7156
    1.6966    2.5961
    1.8351    2.3084
    0.7563   -0.6790
    1.2157    1.5271
   -0.1658   -0.9571
```

```
   -0.1480    2.5377
    1.1049    0.8777
    1.7223    2.4935
clear j i
doc format
format long
pi
ans =
   3.141592653589793
format hex
intmax('uint64')
ans =
   ffffffffffffffff
x = 3;
y = 7;
z1 = x*y;
z2 = x/y;
z3 = x + y
z3 =
   4024000000000000
z4 = mod(7,3);
r = 3;
vol = 4/3 * pi * r^3;
v1 = [10 11 12 13];
v2 = [45 23 12 10];
vsum = v1 + v2
vsum = 404b800000000000   4041000000000000   403800000000000   4037000000000000
format long
v3 = [20; 19; 18];
mv = m * v3
```

```
mv =

   112

   251

   289
vprod = v1 .* v2

vprod =

   450   253   144   130

j1 = 4

j1 =

     4

e = 1/3;

j1sqrt = j1^e

j1sqrt =

   1.587401051968199

j1 = 1:5;

j2 = j1 .^2

j2 =

     1     4     9    16    25

j3 = 1:5;

j4 = (3/2) .^j;

j4 = (3/2) .^j

j4 =  0.918919036439500 + 0.394446199714361i

x = 2;

y = sin(x)

y = 0.909297426825682

xv = [2:.1:3];

yv = sin(xv)

yv = 0.909297426825682   0.863209366648874   0.808496403819590 0.745705212176720
0.675463180551151   0.598472144103956 0.515501371821464   0.427379880233830
0.334988150155905 0.239249329213982   0.141120008059867

zv = exp(xv)
```

zv = 7.389056098930650  8.166169912567650  9.025013499434122 9.974182454814718 11.023176380641601  12.182493960703473 13.463738035001692  14.879731724872837 16.444646771097048 18.174145369443060  20.085536923187668

lv = log(xv)

lv = 0.693147180559945  0.741937344729377  0.788457360364270 0.832909122935104 0.875468737353900  0.916290731874155 0.955511445027436  0.993251773010283 1.029619417181158 1.064710736992428  1.098612288668110

doc ops

doc power

A1 = 1:5;

c = A.^2

c =

    1    16

    4    25

    9    36

c = A1.^2

c =

    1     4     9    16    25

5^5

ans =

       3125

doc mtimes

A1 = [1 1 0 0];

b = [1;2;3;4];

c = a * b

c =

  1.0e+24 *

  0.602300000000000

  1.204600000000000

  1.806900000000000

  2.409200000000000

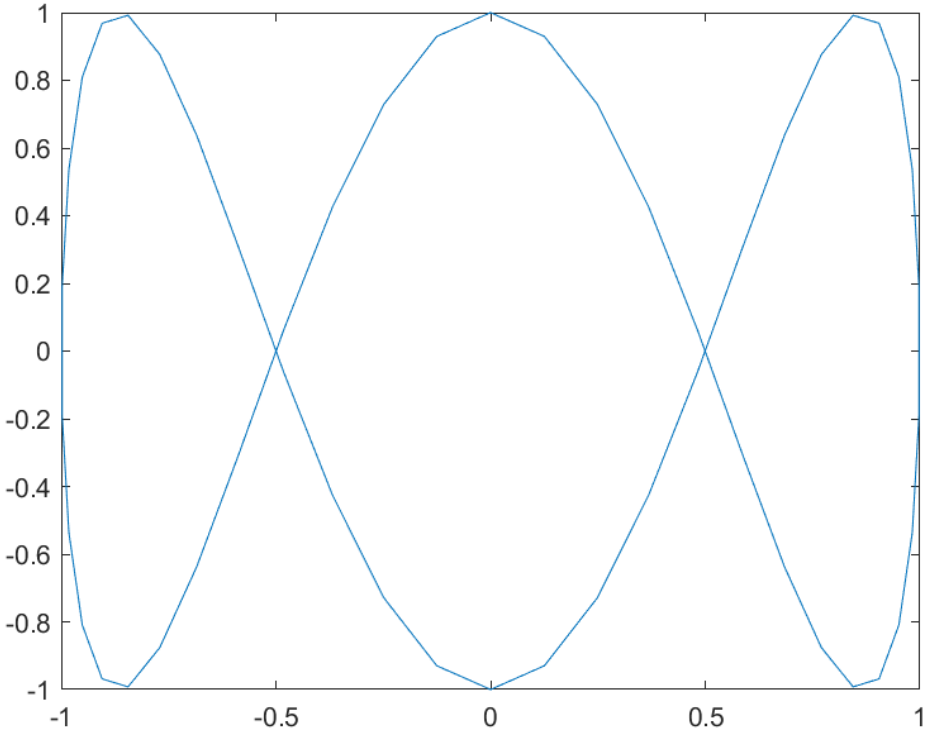
c = A1 * b

c =

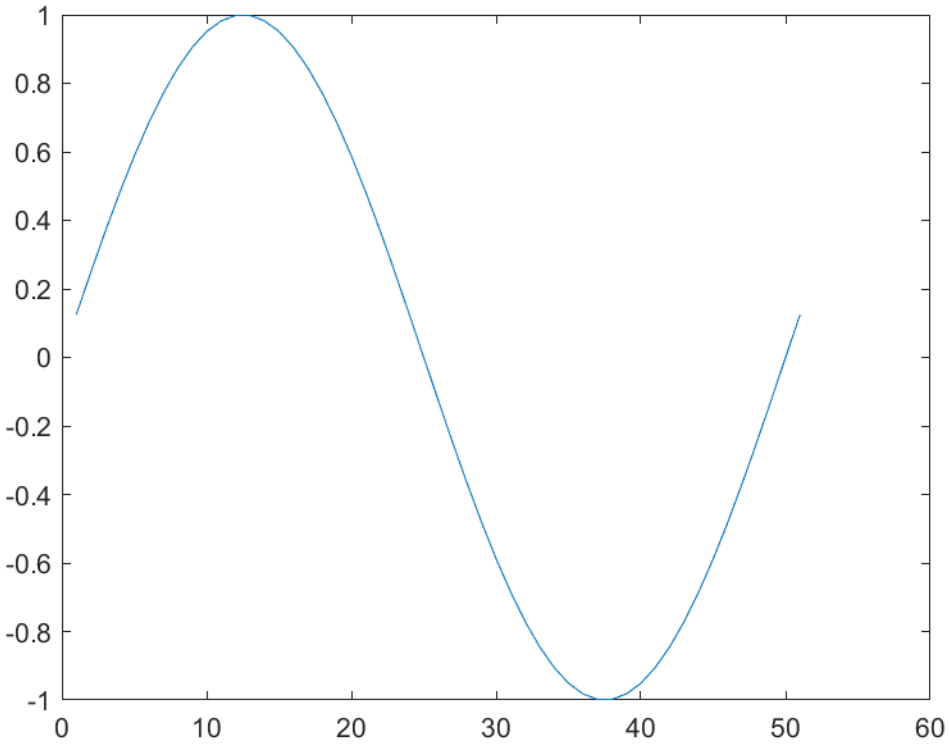
```
     3
c = b * a
c =
   1.0e+24 *
   0.602300000000000
   1.204600000000000
   1.806900000000000
   2.409200000000000
c = b * A1
c =
     1     1     0     0
     2     2     0     0
     3     3     0     0
     4     4     0     0
doc times
a1 = [1 0 3];
b1 = [2 3 7];
c = a1 .* b1
c =
     2     0    21
a1 = 1:3;
b1 = (1:4)';
a1 .* b1
ans =
     1     2     3
     2     4     6
     3     6     9
     4     8    12
doc eig
e = eig(m)
e =
```

```
  10.710426283532634
  -2.408272990792370
   0.697846707259733
[V,D] = eig(m)
V =
   0.345773254739993   0.724508473782354   0.399739782377922
   0.612321198292902  -0.444586429899685  -0.780753609729636
   0.710987768128016  -0.526716600997744   0.480241509324835
D =
  10.710426283532634                   0                   0
                   0  -2.408272990792370                   0
                   0                   0   0.697846707259733
m * V - V*D
ans =
   1.0e-14 *
   0.088817841970013                   0  -0.033306690738755
   0.177635683940025  -0.222044604925031  -0.122124532708767
   0.177635683940025  -0.155431223447522  -0.155431223447522
doc det
DET = det(m)
DET =
   -18
doc inv
m1 =
  -0.277777777777778  -0.500000000000000   0.611111111111111
   0.222222222222222   1.000000000000000  -0.888888888888889
   0.277777777777778  -0.500000000000000   0.388888888888889
b = [4;2;1]
x = m\b
x =
  -1.500000000000000
   2.000000000000000
```

```
   0.500000000000000
doc slash
e = exp(1);
(-sqrt(16)-4)/(2*e^4 + 1)
ans =
  -0.072597718754510
k = (1:20);
k = k.^2;
sum(k)
ans =
        2870
doc prod
temp = [1:3:7; 2:3:8;3:3:9];
temp2 = prod(temp)
temp2 =
    6   120   504
doc cumsum
temp = 1:5;
temp2 = cumsum(temp)
temp2 =
    1     3     6    10    15
doc diff
temp = [1 1 2 3 4 8 13 21];
temp2 = diff(temp)
temp2 =
    0     1     1     1     4     5     8
doc plot
j = (1:51);
x = sin(2*pi*j/50);
y =cos(6*pi*j/50);
plot(x,y)
```

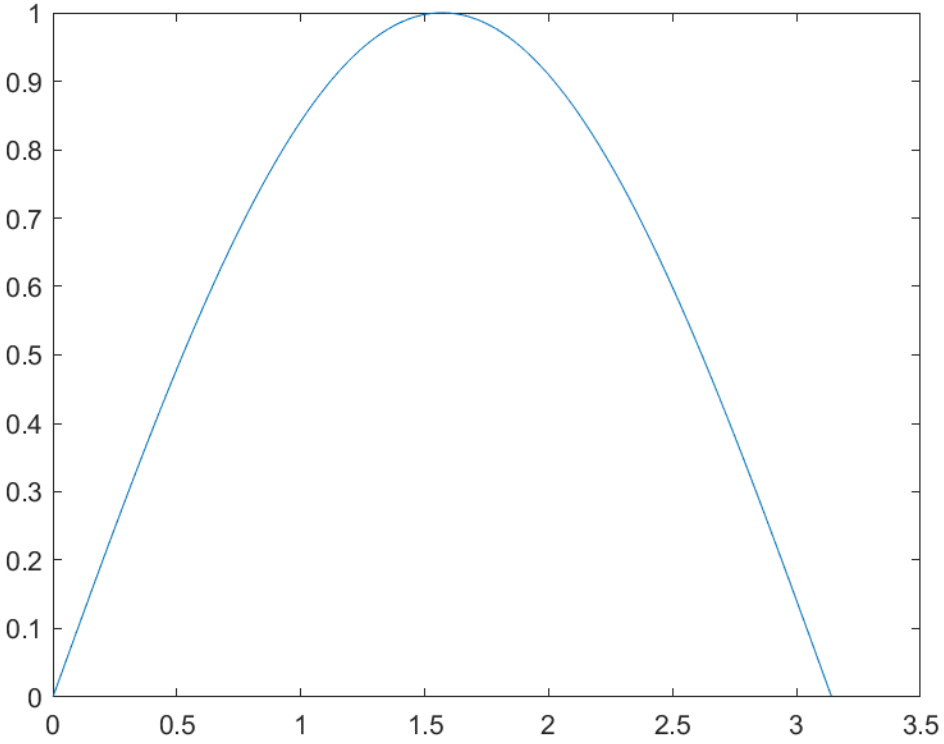



plot(j,x)

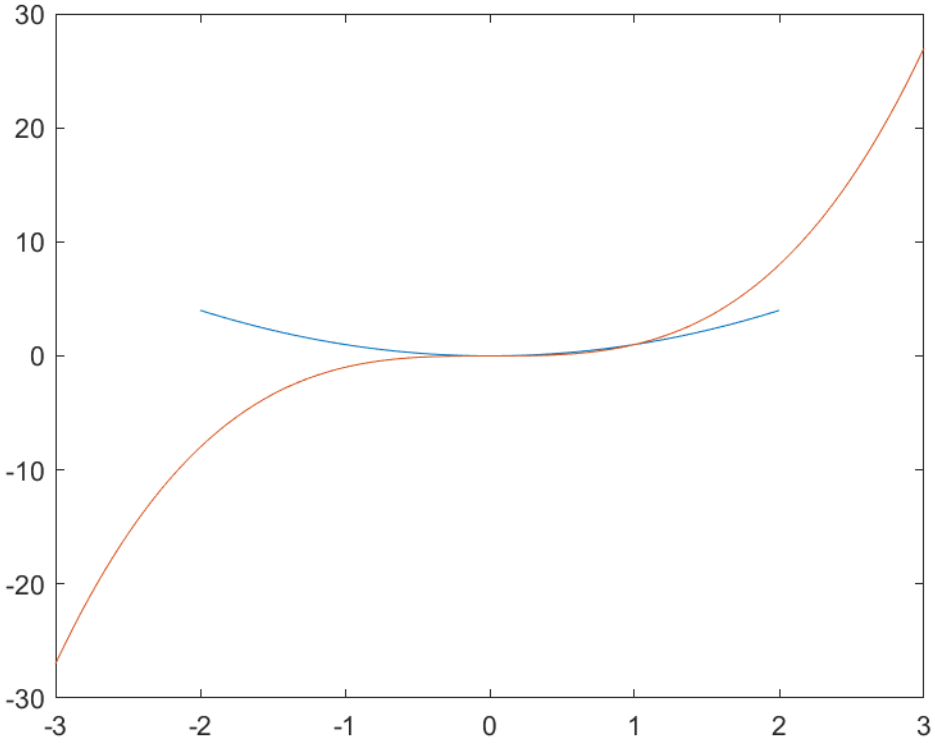



x = 0: .01: pi;

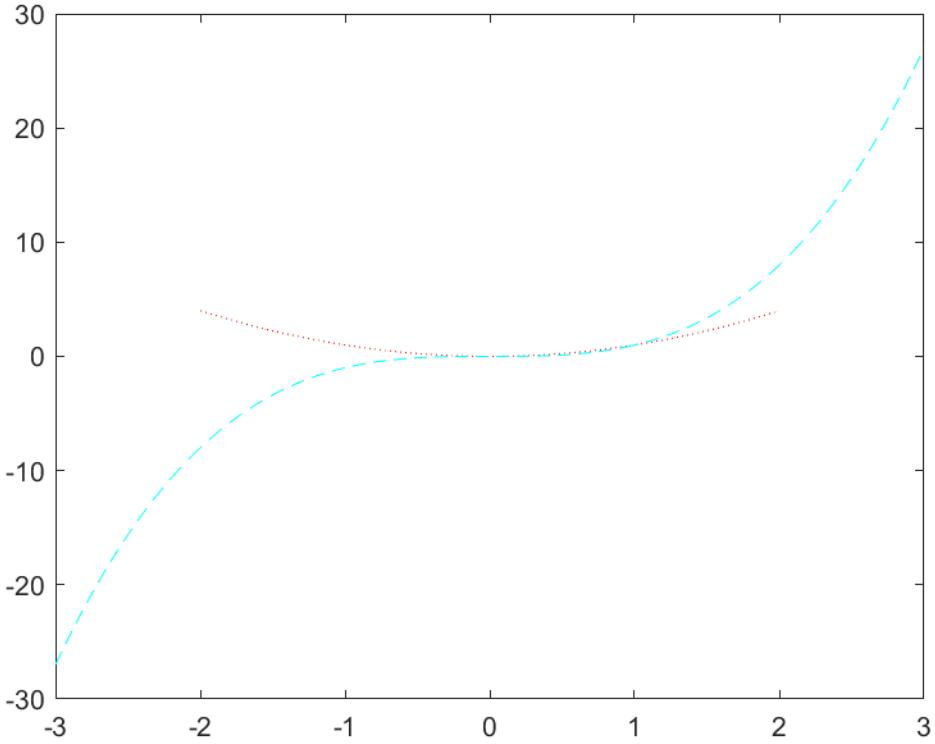
plot(x,sin(x));




```
x1 = -2:.01:2;
y = x1.^2;
x2 = -3:.01:3;
z = x2.^3;
plot(x1,y,x2,z);
```




plot(x1,y,'r:',x2,z,'c--');

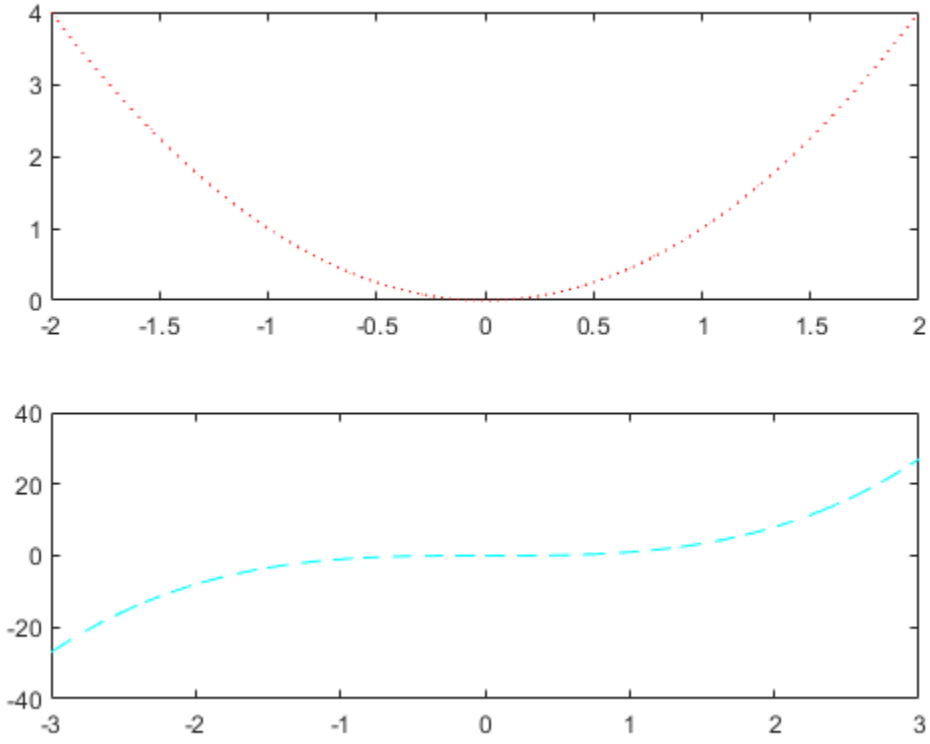



doc subplot

subplot(2,1,1);

plot(x1,y,'r:');

```
subplot(2,1,2);
plot(x2,z,'c--');
```




```
doc figure
f = figure
```

```
f =

  <a href="matlab:helpPopup matlab.ui.Figure" style="font-weight:bold">Figure</a> (3) with properties:

      Number: 3
        Name: ''
       Color: [0.940000000000000 0.940000000000000 … ]
    Position: [403 146 560 420]
       Units: 'pixels'

  Show <a href="matlab:if exist('f', 'var'), matlab.graphics.internal.getForDisplay('f', f, 'matlab.ui.Figure'), else, matlab.graphics.internal.getForDisplay('f'), end">all properties</a>
```

```
f.Position

ans =

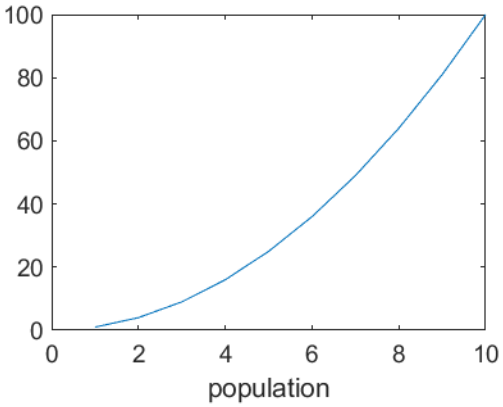
   403   146   560   420
```

f.Position(3:4) = [280 210];

doc xlabel

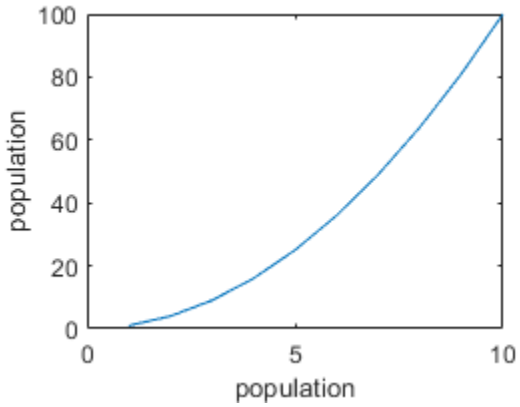
plot((1:10).^2)

xlabel('population');



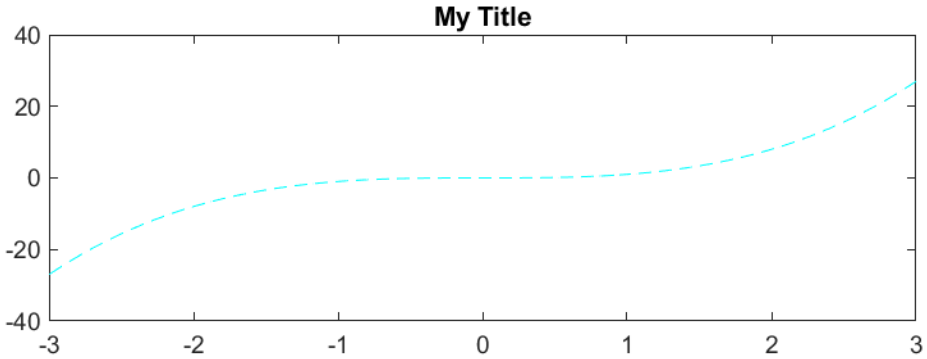

doc ylabel

ylabel('population');




doc title

title('My Title');




doc stem

```
f = figure
```

```
f =
```

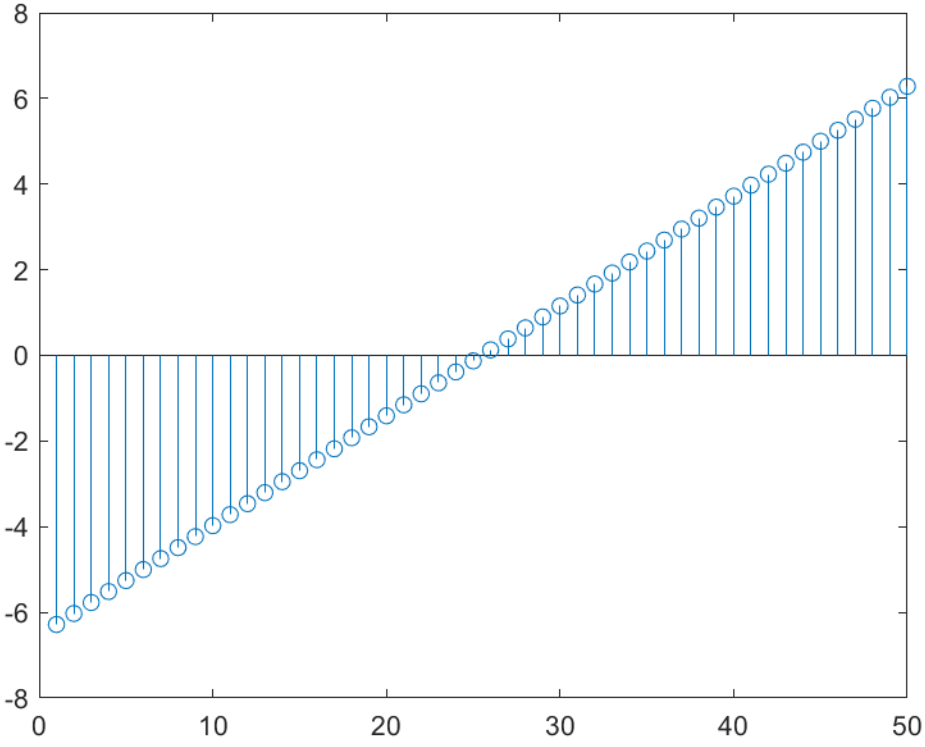
```
  <a href="matlab:helpPopup matlab.ui.Figure" style="font-weight:bold">Figure</a> (2) with properties:

      Number: 2
        Name: ''
       Color: [0.940000000000000 0.940000000000000 … ]
    Position: [403 146 560 420]
       Units: 'pixels'

  Show <a href="matlab:if exist('f', 'var'), matlab.graphics.internal.getForDisplay('f', f, 'matlab.ui.Figure'), else, matlab.graphics.internal.getForDisplay('f'), end">all properties</a>
```

```
Y = linspace(-2*pi,2*pi,50);
```

```
stem(Y)
```

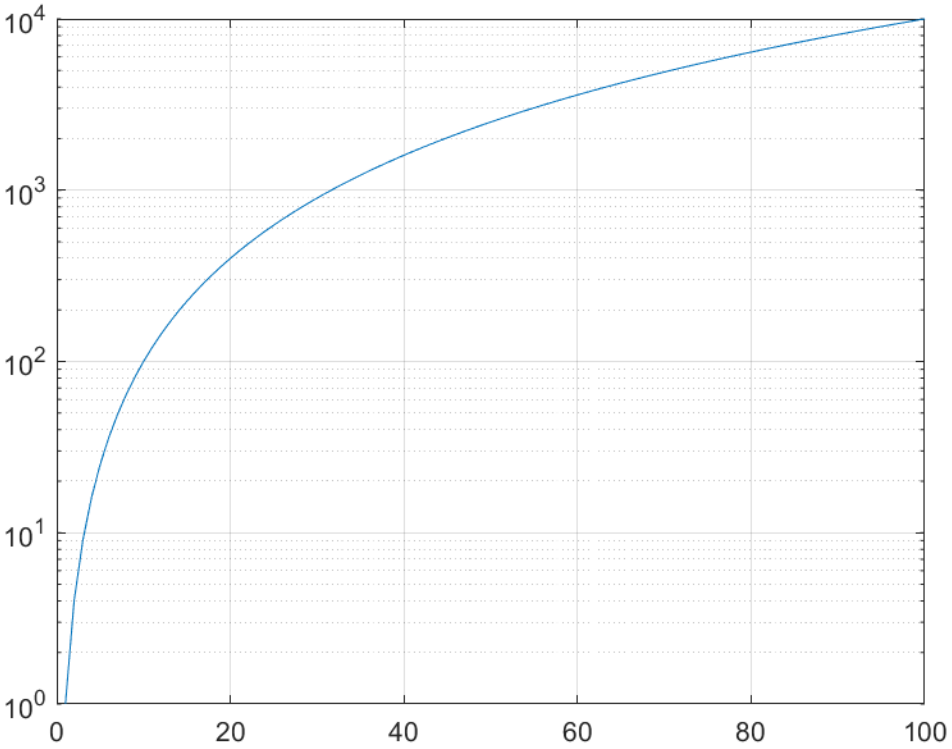



```
doc semilogy
```

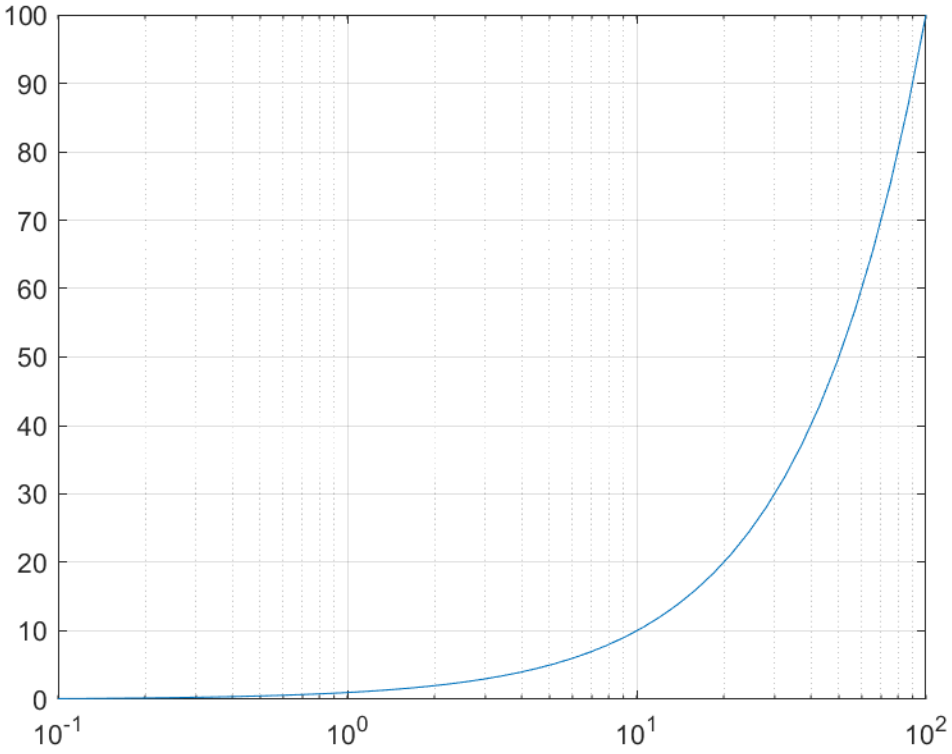
```
x = 1:100;
```

```
y = x.^2;
```

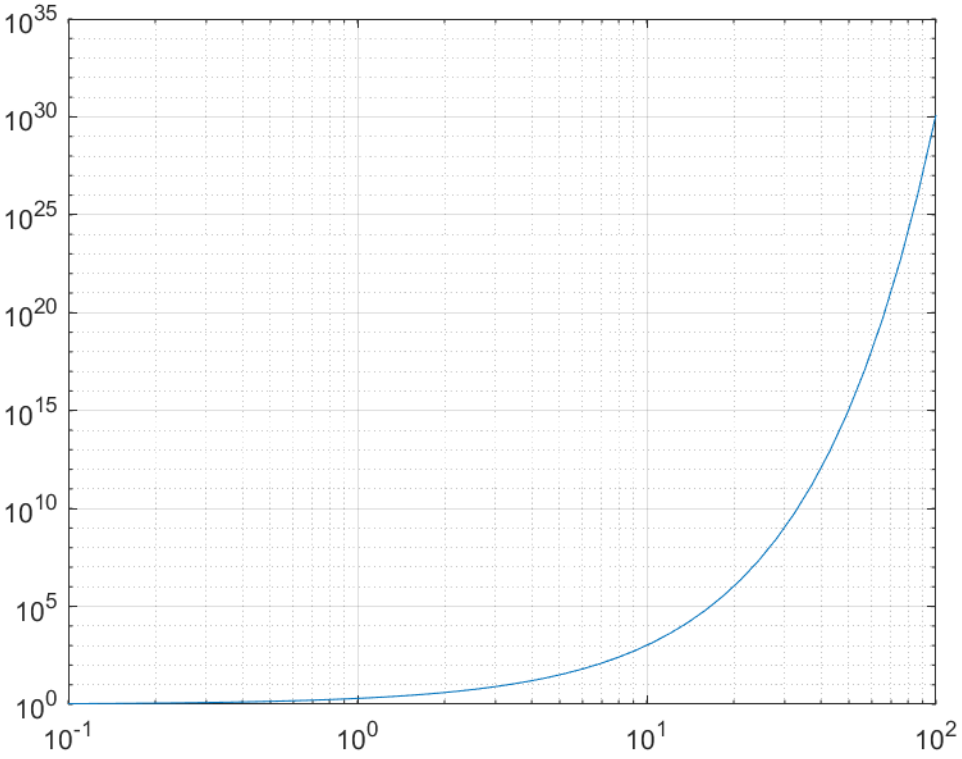
```
semilogy(x,y);
```



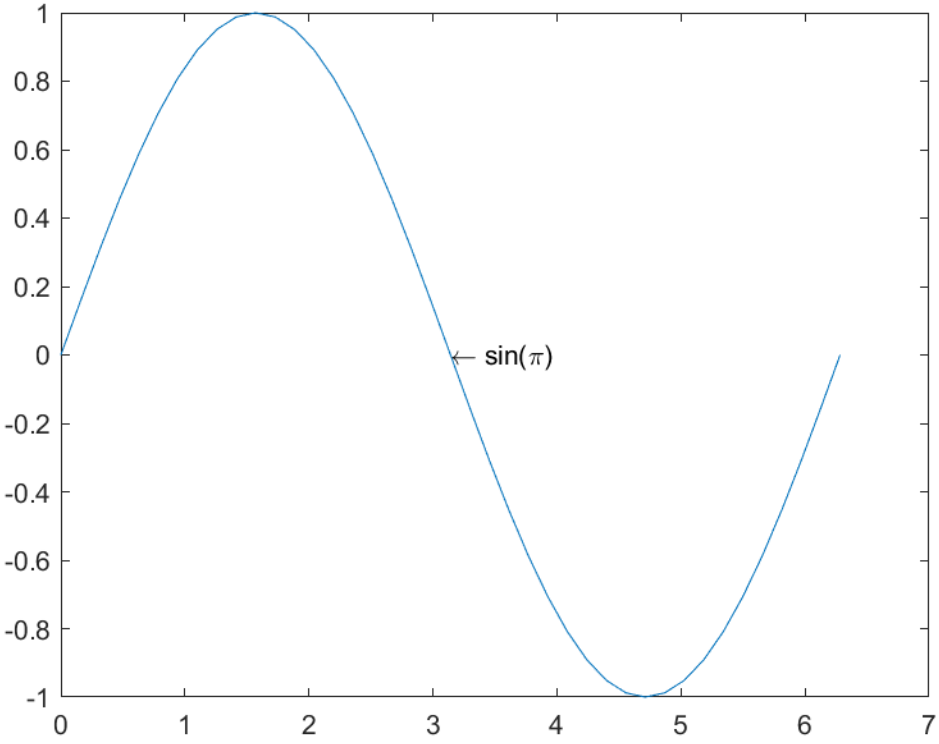

```
grid on
doc semilogx
x = logspace(-1,2);
y = x;
semilogx(x,y)
```




```
grid on
doc loglog
x = logspace(-1,2);
y = 2.^x;
loglog(x,y)
```




```
grid on
doc text
x = 0:pi/20:2*pi;
y = sin(x);
plot(x,y)
text(pi,0,'\leftarrow sin(\pi)')
```




doc print

bar(1:10)

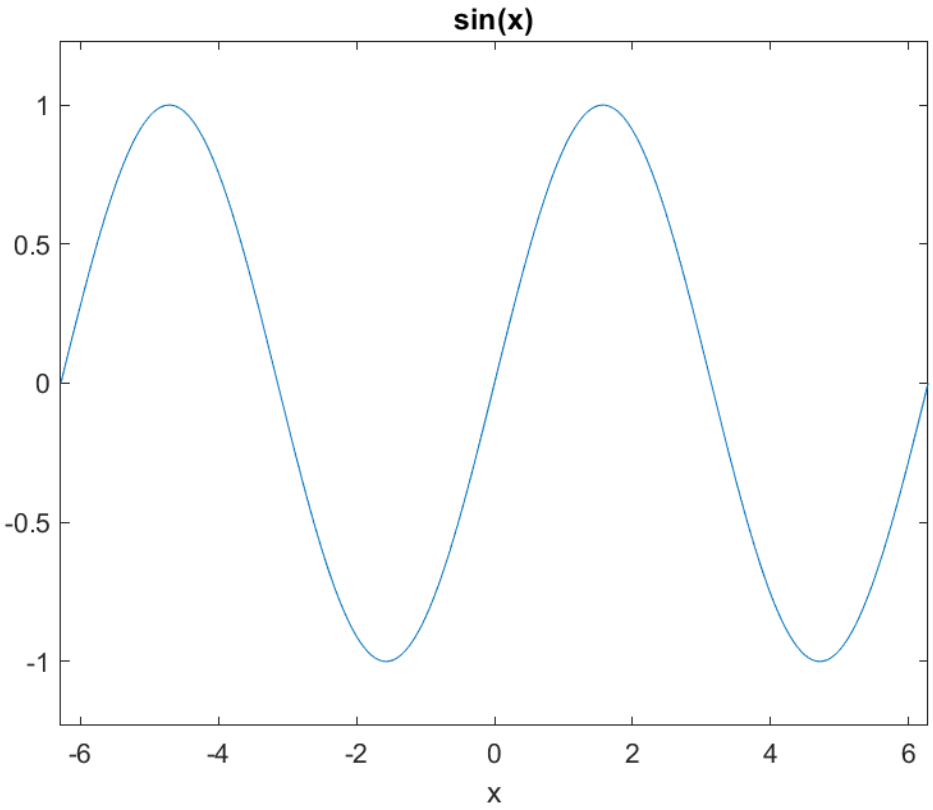
print

**AT END OF DOCUMENT**

doc get

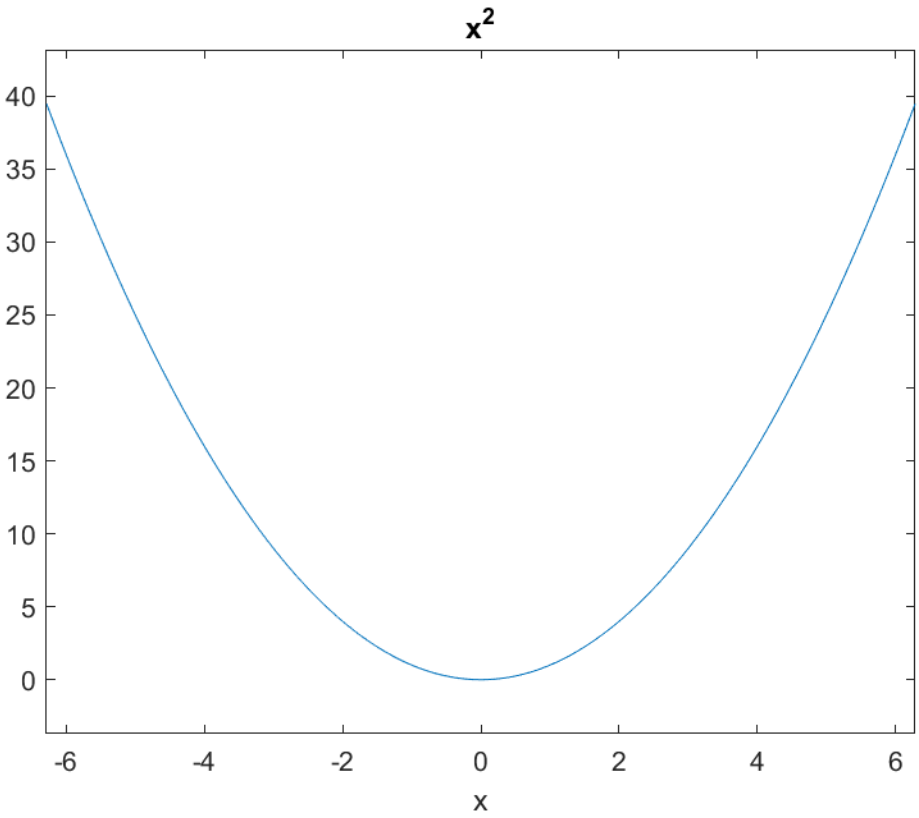
doc set

ezplot('sin(x)')




doc ezplot

ezplot('x^2')




```
help help
lookfor binomial
edit myfile.m
pwd
ans =
    'C:\Users\natew\Documents\continuous\Continuous programs\Matlab_tutorial'
myfile
done!
edit fact
f = fact(4)
f =
    24
fact(5)
ans =
   120
fact(10)
```

```
ans =

    3628800

doc if

edit fact2.m

fact2(10)

ans =

    3628800

doc for

doc while

doc do/while
edit dec2bin.m

edit dectobin.m

b = dectobin(10)

b =

    1     0     1     0
b = dectobin(101)

b =

    1     1     0     0     1     0     1

b = dectobin(15)

b =

    1     1     1     1

b = dectobin(31)

b =

    1     1     1     1     1

doc punct

doc function_handle

f = @(x) sin(x)./x;

f(1)

ans =

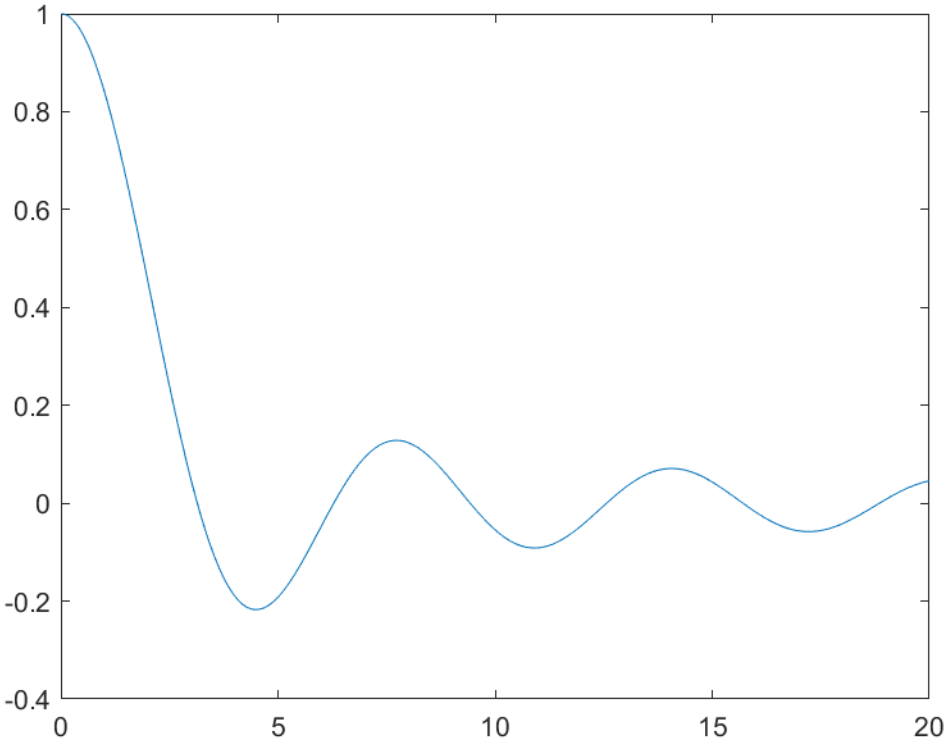
   0.841470984807897

f(2)

ans =
```

0.454648713412841

```
x = .01:.01:20;
plot(x,f(x));
```




```
fact = @(n) prod([1:n]);
fact(4)
```

ans =

24

```
fact(100)
```

ans =

9.332621544394410e+157

```
doc fprintf
a = [1.02 3.04 5.06];
fprintf('%d\n',round(a));
```

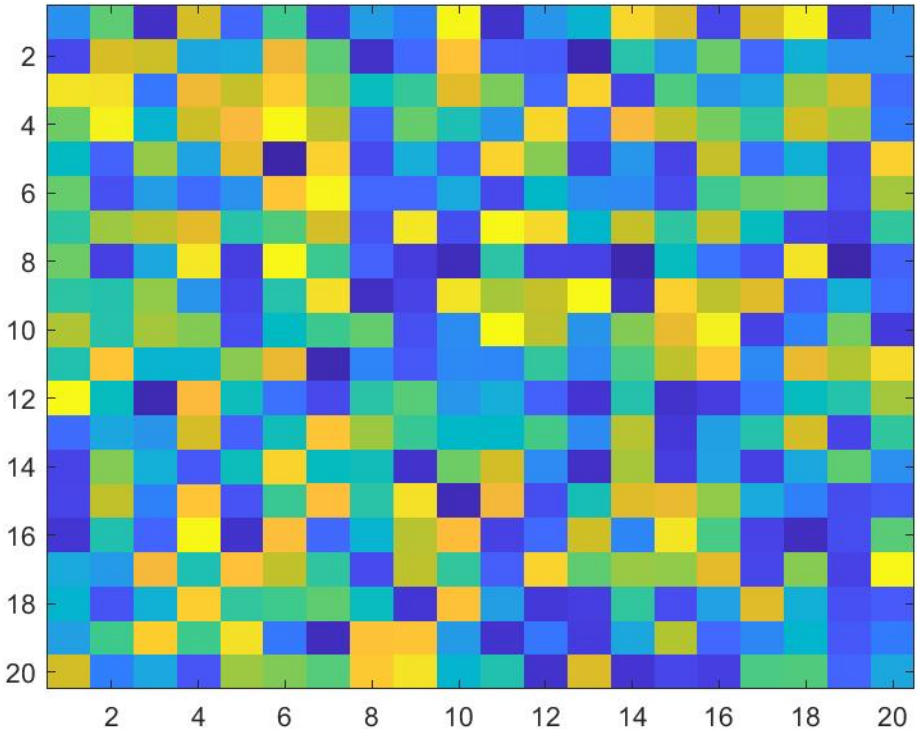
1

3

5

```
doc fopen
```

```
doc fread
doc fclose
ID = fopen('nine.bin','w');
fwrite(ID,[1:9]);
fclose(ID);
ID = fopen('nine.bin');
a = fread(ID)
a =
     1
     2
     3
     4
     5
     6
     7
     8
     9
doc save
doc load
name = "test.mat";
save(name)
clear
load("test.mat")
M = rand(20,20);
imagesc(M);
```




```
fs = 4000;
ts = 1/fs;
t = 0:ts:5;
y = sin(2 * pi * 400 * t);
sound(y,fs);
audiowrite('mysin400.wav',y,fs)
clear y
y = audioread('mysin400.wav');
sound(y,fs)
v = [1, 0, -10, 2]
v =
     1     0   -10     2
rts = roots(v);
v1 = poly(rts)
v1 = 1.000000000000000   0.000000000000002  -9.999999999999998 2.000000000000000
quad('sin',0,5)
ans =
   0.716337813168117
```

```
doc quad
myfun = @(x) 1./(x.^3-2*x-5);
q = quad(myfun,0,2)
q =
  -0.460501739742492
doc symbolic
syms k n;
s1 = symsum(k^2,k,0,n)
s1 =
(n*(2*n + 1)*(n + 1))/6
s2 = expand(s1)
s2 =
n^3/3 + n^2/2 + n/6
subs(s2,n,20)
ans =
2870
syms t T n;
i1 = int(t*cos(2*pi*n*t),t,0,T)
i1 =
-(2*sin(pi*T*n)^2 - 2*T*n*pi*sin(2*pi*T*n))/(4*n^2*pi^2)
simplify(i1)
ans =
-(sin(pi*T*n)^2 - T*n*pi*sin(2*pi*T*n))/(2*n^2*pi^2)
syms e x;
e = (2*x^2 - 3*x + 1)/(x^3 + 2*x^2 - 9*x - 18)
e =
-(2*x^2 - 3*x + 1)/(- x^3 - 2*x^2 + 9*x + 18)
de = diff(e,x)
de =
- (4*x - 3)/(- x^3 - 2*x^2 + 9*x + 18) - ((2*x^2 - 3*x + 1)*(3*x^2 + 4*x - 9))/(- x^3 - 2*x^2 + 9*x + 18)^2
syms d m;
```

```
m = sym([d^2 2 7; d d^3 9; 1 5 1/d])

m =

[d^2,   2,   7]

[  d, d^3,   9]

[  1,   5, 1/d]

det(m)

ans =

d^4 - 7*d^3 - 45*d^2 + 35*d + 16

inv(m)

ans =

[   (d^2 - 45)/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16), (35*d - 2)/(d*(d^4 - 7*d^3 - 45*d^2 + 35*d + 16)),   -(7*d^3 - 18)/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16)]

[           8/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16),        (d - 7)/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16), (- 9*d^2 + 7*d)/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16)]

[(- d^3 + 5*d)/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16),   -(5*d^2 - 2)/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16),  -(-d^5 + 2*d)/(d^4 - 7*d^3 - 45*d^2 + 35*d + 16)]

a = [1 2 -9 -18];

b = [2 -3 1];

[r,p] = residue(b,a)

r =

   0.333333333333334

   4.666666666666665

  -3.000000000000000

p =

   3.000000000000001

  -3.000000000000000

  -2.000000000000000

doc residue

syms x;

f = (2*x^2 - 3*x + 1)/(x^3 + 2*x^2 - 9*x-18);

pfe = diff(int(f))

pfe =
```

```
1/(3*(x - 3)) - 3/(x + 2) + 14/(3*(x + 3))
simplify(pfe)
ans =
1/(3*(x - 3)) - 3/(x + 2) + 14/(3*(x + 3))
num = 2*x^2 - 3 * x + 1
num =
2*x^2 - 3*x + 1
den = x^3 + 2 * x^2 - 9.4*x - 18
den =
x^3 + 2*x^2 - (47*x)/5 - 18
doc deconv
num2 = [2 -3 1];
den2 = [1 2 -9.4 -18];
[q,r] = deconv(num2,den2)
q =
     0
r =
     2    -3     1
syms x
f1 = (2*x^2 - 3*x + 1)/(x^3 + 2*x^2 - 9.4 * x - 18);
diff(int(f1))
ans =
-(5*(2*x^2 - 3*x + 1))/(- 5*x^3 - 10*x^2 + 47*x + 90)
simplify(ans)
ans =
-(5*(2*x^2 - 3*x + 1))/(- 5*x^3 - 10*x^2 + 47*x + 90)
r =
   3.768777065276832
   0.339943600900224
  -2.108720666177055
p =
```

-3.175032623565353

3.039945414333275

-1.864912790767925